FYUGP B.Sc. MATHEMATICS HONOURS

ABSTRACT ALGEBRA I (MAT5CJ302)

Semester V · Core in Major

MODULE 1: BINARY OPERATIONS, GROUPS AND ISOMORPHISMS

Past questions

Section A: 1 to 3 marks

Isomorphic binary structures and group isomorphisms

1 Show that composite of two group isomorphisms is a group isomorphism. **(2 marks)** *November 2022*

2 Let $\phi : G_1 \to G_2$ be an isomorphism of groups. If a has order n in G_1, then show that $\phi(a)$ also has order n in G_2. **(2 marks)** *November 2025*

3 Let $\phi : G_1 \to G_2$ be a group isomorphism. Show that $(\phi(a))^{-1} = \phi\left(a^{-1}\right)$ for all $a \in G$. **(2 marks)** *November 2025*

Groups: definitions and examples

4 The smallest non abelian group has _____ number of elements. **(1 mark)** *November 2021*

5 Let G be a nonempty set with an associative binary operation in which the equations $ax = b$ and $xa = b$ have solutions for all $a, b \in G$. Prove that G is a group. **(2 marks)** *November 2024*

Elementary properties of groups

6 The order of the identity element in any group G is _____. **(1 mark)** *November 2021*

7 Let G be a group and $a, b \in G$. Show that $(ab)^{-1} = b^{-1}a^{-1}$. **(3 marks)** *November 2021*

8 Show that cancellation property holds in a group G. **(2 marks)** *November 2022*

9 State and prove cancellation property for groups. **(2 marks)** *November 2023*

10 Let G be a group and $a, b, c \in G$. Prove that if $ab = ac$ then $b = c$. **(2 marks)** *November 2023*

11 Let G be group. Prove that G is abelian if and only if $(ab)^{-1} = a^{-1}b^{-1}$ for all $a, b \in G$. **(2 marks)** *November 2024*

12 Show that in a group G and $a, b \in G$, each of the equations $ax = b$ and $xa = b$ has a unique solution. **(2 marks)** *November 2025*

Finite groups and group tables

13 Give addition table for $\mathbb{Z}_2 \times \mathbb{Z}_2$. **(2 marks)** *November 2022*

14 Make multiplication table for $\mathbb{Z}_7$. **(2 marks)** *November 2023*

15 Write the addition table for $\mathbb{Z}_8$. **(2 marks)** *November 2025*

Section B: 4 to 7 marks

Binary operations

16 Let S be a set and let f, g and h be functions mapping S into S. Prove that $(f \circ g) \circ h = f \circ (g \circ h)$. **(4 marks)** *November 2018*

17 Define $*$ on $\mathbb{Z}$ by $a * b = a - b$. Check whether $(\mathbb{Z}, *)$ is a group. **(6 marks)** *November 2021*

Isomorphic binary structures and group isomorphisms

18 Show that the binary structure $\langle \mathbb{R}, + \rangle$ with operation the usual addition is isomorphic to the structure $\langle \mathbb{R}^+, \cdot \rangle$ where $\cdot$ is the usual multiplication. **(7 marks)** *November 2018*

19 Let G and G' be groups and let $\Phi : G \to G'$ be one to one function such that $\Phi(xy) = \Phi(x)\Phi(y)$ for all $x, y \in G$. Then prove that $\Phi[G]$ is a subgroup of G' and Φ provides an isomorphism of G with $\Phi[G']$. **(7 marks)** *November 2018*

20 Consider the following two binary structures : (a) $\mathbb{Z}$, the set of integers with ordinary addition ; and (b) $2\mathbb{Z} = \{2n \mid n \in \mathbb{Z}\}$ the set of even integers with ordinary addition. Show that the above two binary structures are isomorphic. **(4 marks)** *November 2021*

21 Let $\varphi : G \to H$ be an isomorphism of groups. Then show that : a) If G is abelian then so is H ; and b) If G is cyclic then so is H. **(5 marks)** *November 2022*

Groups: definitions and examples

22 Define $*$ by $a * b = \dfrac{ab}{3}$ then show that $\mathbb{Q}^+$, the set of positive rationals, with operation $*$ is a group. **(4 marks)** *November 2018*

23 Show that the subset S of $M_n(\mathbb{R})$ consisting of all invertible $n \times n$ matrices under matrix multiplication is a group. **(7 marks)** *November 2018*

24 Let $*$ be defined by $\mathbb{Q}^+$ by $a * b = \dfrac{ab}{2}$. Show that $(\mathbb{Q}^+, *)$ is an abelian group. **(7 marks)** *November 2021*

25 Let G_1 and G_2 be groups. Prove that the direct product $G_1 \times G_2$ is a group under the operation defined for all $(a_1, a_2), (b_1, b_2) \in G_1 \times G_2$ by $(a_1, a_2)(b_1, b_2) = (a_1b_1, a_2b_2)$. **(5 marks)** *November 2023*

26 Prove that the units of $\mathbb{Z}_8$ forms a group under multiplication of congruences. **(5 marks)** *November 2024*

27 Let $S = \mathbb{R} - \{-1\}$. Define $*$ on S by $a * b = a + b + ab$. Show that $(S, *)$ is a group. **(5 marks)** *November 2025*

Elementary properties of groups

28 Show that left and right cancellation holds in a group G. **(4 marks)** *November 2021*

29 Let G be a group and suppose that $a * b * c = e \; \forall a, b, c \in G$. Show that $b * c * a = e$. **(4 marks)** *November 2021*

30 Prove that a group G has exactly one idempotent element. **(4 marks)** *November 2021*

31 Let G be a group. For all $a, b \in G$, prove that $(a * b)^{-1} = b^{-1} * a^{-1}$. **(7 marks)** *November 2021*

32 Let G be a group. Prove that if $x^2 = e$ for all $x \in G$ then G is abelian. **(5 marks)** *November 2022*

33 Let G be a group and $a, b \in G$. Prove that $(ab)^n = a^n b^n$ for all $n \in \mathbb{Z}$ if and only if $ab = ba$. **(5 marks)** *November 2022*

34 If G is a group and $a, b \in G$, then prove that each of the equations $ax = b$ and $xa = b$ has a unique solution. **(5 marks)** *November 2023*

Finite groups and group tables

35 Show that every group of prime order is abelian. **(4 marks)** *November 2018*

36 Draw the group table for S_3. **(4 marks)** *November 2018*

37 Let n be a positive integer. Give an example of a group containing n elements. **(4 marks)** *November 2021*

Section C: 10 marks

Isomorphic binary structures and group isomorphisms

38 Show that the inverse of a group isomorphism is a group isomorphism. **(10 marks)** *November 2022*

39 a) Let $\phi : G_1 \to G_2$ be an isomorphism of groups. Prove that ϕ preserves following structural properties : (i) If a has order n in G_1, then $\phi(a)$ has order n in G_2, (ii) If G_1 is abelian, then so is G_2, (iii) If G_1 is cyclic, then so is G_2. b) Prove that $\mathbb{Z}_4 \ncong \mathbb{Z}_2 \times \mathbb{Z}_2$. **(10 marks)** *November 2024*